

LECTURE SHEET

EXERCISE-I

(Theories of Acids-Bases)

LEVEL-I (MAIN)

Straight Objective Type Questions

- Which of the following is an Arrhenius acid?
1) NH_3 2) SO_2 3) AlCl_3 4) HNO_3
- The conjugate base of hydrazoic acid is
1) N^{3-} 2) N_3^- 3) NH_2^- 4) NH_4^+
- The conjugate base of H_2PO_4^- is
1) H_3PO_4 2) P_2O_5 3) PO_4^{3-} 4) HPO_4^{2-}
- O_2^{2-} is the conjugate base of
1) OH^- 2) H_2O 3) H_2O_2 4) HO_2^-
- Which of the following can act as Lowry-Bronsted acid as well as base?
1) HCl 2) SO_4^{2-} 3) HPO_4^{2-} 4) Br^-
- Which of the following is an acidic salt ?
1) Na_3PO_4 2) Na_2HPO_4 3) NaH_2PO_2 4) NaH_2PO_4
- Which is the correct order of decreasing basicity of halide ions?
1) $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$ 2) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$ 3) $\text{Cl}^- > \text{Br}^- > \text{I}^- > \text{F}^-$ 4) $\text{F}^- > \text{I}^- > \text{Br}^- > \text{Cl}^-$
- Which of the following is a Bronsted neutralisation reaction ?
1) $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$ 2) $\text{HCl} + \text{NH}_3 \rightarrow \text{NH}_4^+ + \text{Cl}^-$
3) $\text{NH}_3 + \text{BF}_3 \rightarrow [\text{H}_3\text{N} \rightarrow \text{BF}_3]$ 4) $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
- The solution of acetic acid in benzene contains
1) CH_3COO^- 2) H^+ 3) H_3O^+ 4) $(\text{CH}_3\text{COOH})_2$
- Which of the following is a Lewis acid ?
1) CCl_4 2) BF_3 3) H_2O 4) SO_4^{2-}
- Which of the following acts as Lewis base ?
1) C_2H_2 2) C_2H_4 3) Pyridine 4) All the above
- Which of the following acts as Lewis acid ?
1) H 2) He 3) S 4) B
- Which of the following species acts as a Lewis acid and also as a Lewis base ?
1) SO_2 2) SCl_4 3) both SO_2 and SCl_4 4) SO_3
- In the reaction, $\text{AlCl}_3 + \text{Cl}^- \rightarrow \text{AlCl}_4^-$, Cl^- acts as
1) Bronsted acid 2) Bronsted base 3) Lewis base 4) Lewis acid

15. Which of the following failed to explain the relative strength of acids and bases ?
 1) Arrhenius theory 2) Bronsted theory 3) Lowry theory 4) Lewis theory
16. Al^{3+} is a Lewis acid, because
 1) It has vacant orbital 2) It has completely filled orbitals
 3) It can give H^+ 4) It can add with OH^-
17. Which of the following has least tendency to act as Lewis acid ?
 1) I^- 2) I^+ 3) $SnCl_2$ 4) $AlCl_3$

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. Which of the following does not turn blue litmus to red ?
 a) 1M HCl b) Pure and dry HCl c) dil.HCl d) Conc.HCl
2. Conjugate base of H_2 is
 a) Electron b) Hydride ion c) Proton d) Neutron
3. In the reaction, $CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$, H_3O^+ acts as
 a) Weak acid b) Weak base c) Strong acid d) Strong base
4. The no. of conjugate acid-base pairs present in the aqueous solution of H_3PO_3 is
 a) 2 b) 3 c) 4 d) 5
5. Which of the following is a Bronsted acid but not a Bronsted base ?
 a) H_2O b) NH_3 c) H_2S d) HCO_3^-
6. The conjugate base of $[Al(H_2O)_3(OH)_3]$ is
 a) $[Al(H_2O)_3(OH)_2]^+$ b) $[Al(H_2O)_3(OH)_2O]^-$ c) $[Al(H_2O)_3(OH)_3]^+$ d) $[Al(H_2O)_2(OH)_4]^-$
7. In aqueous solution, HCl and HNO_3 are equally strong. This is because
 a) Their basicities are same b) Both are oxy acids of non-metals
 c) Both have lower molecular weights d) Levelling effect of water
8. Among the following, the stronger acid in basic medium is
 a) Aniline b) Phenol c) Ammonia d) Benzene
9. Assertion (A) : H_3O^+ is the strongest acid in aqueous solution.
 Reason (R) : water levels the strength of hydronium ion.
 a) Both (A) and (R) are true and (R) is the correct explanation of (A)
 b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 c) (A) is true but (R) is false d) (A) is false but (R) is true
10. Which of the following is strong Lewis base?
 a) NF_3 b) NCl_3 c) NBr_3 d) NI_3
11. Which of the following is a relatively strong Lewis acid ?
 a) BF_3 b) BCl_3 c) BBr_3 d) BI_3
12. Which of the following acts as a Lewis acid in the following reaction $SnCl_4 + 2Cl^- \rightarrow [SnCl_6]^{2-}$
 a) Cl^- b) $[SnCl_6]^{2-}$ c) $SnCl_4$ d) $2Cl^-$

More than One correct answer Type Questions

13. Which of the following species can be formed in acid - base reactions as Bronsted bases as well as Bronsted acids?
- a) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ b) HPO_4^{2-} c) HSO_4^- d) NH_4^+
14. In the following reaction :
- $$[\text{Cu}(\text{H}_2\text{O})_3(\text{OH})]^+ + [\text{Al}(\text{H}_2\text{O})_6]^{3+} \rightarrow [\text{Cu}(\text{H}_2\text{O})_4]^{2+} + [\text{Al}(\text{H}_2\text{O})_5(\text{OH})]^{2+}$$
- (A) (B) (C) (D)
- a) A is an acid and B is the base b) A is a base and B is the acid
 c) C is the conjugate acid of A and D is the conjugate base of B
 d) C is conjugate base of A and D is the conjugate acid of B
15. Consider the equilibrium equation : $\text{CH}_3\text{COOH} + \text{HF} \rightleftharpoons \text{CH}_3\text{COOH}_2^+ + \text{F}^-$
 Which of the following statement(s) is/are incorrect:
- a) F^- is the conjugate base of HF
 b) F^- is the conjugate base of CH_3COOH
 c) $\text{CH}_3\text{COOH}_2^+$ is the conjugate base of CH_3COOH
 d) CH_3COOH is the conjugate base of $\text{CH}_3\text{COOH}_2^+$
16. Which of the following statements are correct ?
- a) OH^- is a stronger base than NH_2^-
 b) H_2O is a stronger base than CH_3OH
 c) CH_3COO^- is a stronger base than HCOO^-
 d) A Lewis acid is a compound that accepts a pair of electrons.
17. The equilibrium constant for the reaction $\text{HNO}_{2(aq)} + \text{CN}^-_{(aq)} \rightleftharpoons \text{HCN}_{(aq)} + \text{NO}_2^-_{(aq)}$ is 1.1×10^6 . From the magnitude of this K_{eq} one can conclude that
- a) CN^- is stronger base than NO_2^- b) HCN is a stronger acid than HNO_2
 c) The conjugate base of HNO_2 is NO_2^- d) The conjugate acid of CN^- is HCN
18. Which of the following statements are correct:
- a) Arrhenius acids are also Bronsted acids but all Arrhenius bases are not Bronsted bases
 b) All Bronsted bases are also Lewis bases
 c) All Bronsted acids are also Lewis acids
 d) A strong acid has a weak conjugate base but a strong base has a strong conjugate acid
19. Which of the following has been arranged correctly in increasing order of acidic strength?
- a) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$ b) $\text{H}_3\text{PO}_2 < \text{H}_3\text{PO}_3 < \text{H}_3\text{PO}_4$
 c) $\text{H}_3\text{PO}_4 < \text{H}_2\text{SO}_4 < \text{HClO}_4$ d) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{BiH}_3$

Matrix Matching Type Questions

20. Column-I

- A) CCl_4
 B) AlCl_3
 C) NH_3
 D) HClO_4

Column-II

- P) Lewis acid
 Q) Lewis base
 R) Bronsted acid
 S) Bronsted base
 T) Neither lewis acid nor Lewis base

21. Column-I

- A) HPO_4^{-2}
 B) CO_3^{-2}
 C) $\text{H}_2\text{PO}_3^{-1}$
 D) $\text{H}_2\text{PO}_2^{-1}$

Column-II

(In the aqueous medium the ion can be)

- P) Arrhenius acid
 Q) Bronsted base
 R) Amphoteric
 S) Bronsted acid

Integer Type Questions

22. How many of the following are lewis acids? $[\text{CCl}_4, \text{SiF}_4, \text{BF}_4^-, \text{NH}_4^+, \ddot{\text{C}}\text{Cl}_2, \text{Mg}^{+2}, \text{O}, \text{HCHO}]$

EXERCISE-II

(Ostwald's dilution law, ionic product of water, K_a , K_b , pH concept for acids and bases)

LEVEL-I (MAIN)
Straight Objective Type Questions

- Dissociation constant of water at 25°C is
 1) 1.0×10^{-14} 2) 1×10^{14} 3) 14 4) 1.8×10^{-16}
- At any temperature, $\text{pH} + \text{pOH}$ is equal to
 1) 7 2) 0 3) 14 4) $\text{p}K_w$
- At any temperature, the proton concentration of water is
 1) 10^{-7}M 2) $< 10^{-7}\text{M}$ 3) $> 10^{-7}\text{M}$ 4) $\sqrt{K_w}$
- At a certain temperature, the H^+ ion concentration of water is $4 \times 10^{-7}\text{M}$ then the value of K_w at the same temperature is
 1) 10^{-14}M^2 2) $4 \times 10^{-14}\text{M}^2$ 3) $1.6 \times 10^{-13}\text{M}^2$ 4) $4 \times 10^{-7}\text{M}^2$
- At 298 K, the $[\text{H}_3\text{O}^+]$ of a solution is $2 \times 10^{-9}\text{M}$. The nature of the solution is
 1) Acidic 2) Basic 3) Neutral 4) Cannot be predicted
- At a given temperature, When an acid is added to water then the value of K_w
 1) Decreases 2) Increases
 3) Remains same 4) First decreases then increases.
- At 100°C , the pH of pure water is
 1) 7 2) Greater than 7 3) Less than 7 4) Zero
- At some high temperature, K_w of water is 10^{-13} . Then the pH of the water at the same temperature is
 1) 7.0 2) 6.5 3) 7.5 4) 7.23
- Among the following
 a) pH of water decreases with increase in temperature
 b) pH of water decreases by the addition of base
 c) pH of water increases by the addition of acid
 d) At any temperature pH of water is equal to $\text{p}K_w / 2$
 1) All are correct 2) b, c, d are correct
 3) a and d are correct 4) a, b and c are correct

21. Column-I

- A) HPO_4^{-2}
 B) CO_3^{-2}
 C) $\text{H}_2\text{PO}_3^{-1}$
 D) $\text{H}_2\text{PO}_2^{-1}$

Column-II

(In the aqueous medium the ion can be)

- P) Arrhenius acid
 Q) Bronsted base
 R) Amphoteric
 S) Bronsted acid

Integer Type Questions

22. How many of the following are lewis acids? $[\text{CCl}_4, \text{SiF}_4, \text{BF}_4^-, \text{NH}_4^+, \ddot{\text{C}}\text{Cl}_2, \text{Mg}^{+2}, \text{O}, \text{HCHO}]$

EXERCISE-II

(Ostwald's dilution law, ionic product of water, K_a , K_b , pH concept for acids and bases)

LEVEL-I (MAIN)
Straight Objective Type Questions

- Dissociation constant of water at 25°C is
 1) 1.0×10^{-14} 2) 1×10^{14} 3) 14 4) 1.8×10^{-16}
- At any temperature, $\text{pH} + \text{pOH}$ is equal to
 1) 7 2) 0 3) 14 4) $\text{p}K_w$
- At any temperature, the proton concentration of water is
 1) 10^{-7}M 2) $< 10^{-7}\text{M}$ 3) $> 10^{-7}\text{M}$ 4) $\sqrt{K_w}$
- At a certain temperature, the H^+ ion concentration of water is $4 \times 10^{-7}\text{M}$ then the value of K_w at the same temperature is
 1) 10^{-14}M^2 2) $4 \times 10^{-14}\text{M}^2$ 3) $1.6 \times 10^{-13}\text{M}^2$ 4) $4 \times 10^{-7}\text{M}^2$
- At 298 K, the $[\text{H}_3\text{O}^+]$ of a solution is $2 \times 10^{-9}\text{M}$. The nature of the solution is
 1) Acidic 2) Basic 3) Neutral 4) Cannot be predicted
- At a given temperature, When an acid is added to water then the value of K_w
 1) Decreases 2) Increases
 3) Remains same 4) First decreases then increases.
- At 100°C , the pH of pure water is
 1) 7 2) Greater than 7 3) Less than 7 4) Zero
- At some high temperature, K_w of water is 10^{-13} . Then the pH of the water at the same temperature is
 1) 7.0 2) 6.5 3) 7.5 4) 7.23
- Among the following
 a) pH of water decreases with increase in temperature
 b) pH of water decreases by the addition of base
 c) pH of water increases by the addition of acid
 d) At any temperature pH of water is equal to $\text{p}K_w / 2$
 1) All are correct 2) b, c, d are correct
 3) a and d are correct 4) a, b and c are correct

10. The H^+ ion concentration of a solution is 4×10^{-5} M. Then the OH^- ion concentration of the same solution is
 1) 4×10^{-5} M 2) 2.5×10^{-9} M 3) 1.0×10^{-7} M 4) 2.5×10^{-10} M
11. 0.004g of NaOH is present in 1L. of a solution. The pH of the solution is
 1) 14.0 2) 12.0 3) 10.0 4) 8.0
12. The P^H of 0.005 M $Ba(OH)_2$ is
 1) 2.301 2) 11.699 3) 12 4) 7
13. The $[OH^-]$ of 0.005 M H_2SO_4 is
 1) 2×10^{-12} M 2) 5×10^{-3} M 3) 10^{-2} M 4) 10^{-12} M
14. The P^H of a dibasic acid is 3.699. Its molarity is
 1) 2×10^{-4} M 2) 4×10^{-4} M 3) 2×10^{-3} M 4) 1×10^{-4} M
15. The P^H of 10^{-8} M HCl is
 1) 8 2) 6 3) 7 4) 6.98
16. The P^H of 0.1M NaCl solution is
 1) 1 2) 13 3) 7 4) Zero
17. The P^H of a solution is 3.0. This solution is diluted by 100 times. Then the P^H of the resulting solution is
 1) 5 2) 7 3) 1 4) 11
18. The P^H of HCl is 3. Then the P^H of NaOH solution having same molar concentration is
 1) 3 2) 6 3) 9 4) 11
19. For a weak acid (α is very small)
 1) $K_a = C \cdot \alpha^2$ 2) $\alpha = \sqrt{\frac{K_a}{C}}$ 3) $[H^+] = C \cdot \alpha$ 4) all the above
20. At infinite dilution, the percentage ionisation of both strong and weak electrolytes is
 1) 1 % 2) 20 % 3) 50 % 4) 100 %
21. Which of the following is the best conductor of electricity ?
 1) 1M HNO_3 2) 1M H_2CO_3 3) 1M H_3PO_4 4) 1M H_2SO_4
22. Strength of a weak acid or a weak base depends upon its
 1) Temperature 2) Nature of solvent
 3) Degree of dissociation 4) All the above
23. The P^H of 0.001 M CH_3COOH is
 1) 3 2) 11 3) Between 3 and 7 4) 7
24. Which of the following is relatively stronger acid? K_a values are given in brackets
 1) HA (2×10^{-4}) 2) HB (3×10^{-5}) 3) HC (1.8×10^{-3}) 4) HD (9.6×10^{-10})
25. A monoprotic acid in a 0.1 M solution ionizes to 0.001%. Its ionization constant is
 1) 1.0×10^{-3} 2) 1.0×10^{-6} 3) 1.0×10^{-8} 4) 1.0×10^{-11}
26. The hydrogen ion concentration of 0.2M CH_3COOH which is 4% ionised is
 1) 0.008 N 2) 0.12 N 3) 0.8 N 4) 0.08 N
27. pH of a weak acid solution in 0.1 M solution is 4.3. What is the pH in 0.01 M solution ?
 1) 4.1 2) 4.8 3) 5.3 4) 7.3

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- Liquid ammonia ionises to a slight extent. Its self ionisation constant is 10^{-30} M^2 at -50°C , then the number of amide ions present per 1 cc of it at -50°C is
a) 6×10^{12} b) 3×10^{12} c) 6×10^5 d) 3×10^3
- The P^{H} of a solution is 6. Its $[\text{H}_3\text{O}^+]$ is decreased by 1000 times. Its P^{H} will be
a) 9 b) 6.96 c) 7.04 d) 8
- The P^{H} of a solution is 11. It is diluted by 1000 times. Then the P^{H} of resulting solution is
a) 8 b) 14 c) 7 d) 7.04
- The P^{H} of 40 ml of 0.02 M HCl will not be changed by adding
a) 1 ml of 1M HCl b) 2 ml of 1M NaOH
c) 20 ml of 0.1 M NaCl d) 36 ml of same concentrated HCl solution
- The P^{H} of HCl is 1. The amount of NaOH to be added to 100 ml of such a HCl solution to get P^{H} of 7 is
a) 4g b) 0.4g c) 4 mg d) 0.4 mg
- How many grams of NaOH can neutralise 100 ml of H_2SO_4 solution with pH value 3.301 ?
a) 2 mg b) 3 mg c) 4 mg d) 8 mg
- 20 ml of 0.4 M H_2SO_4 and 80 ml of 0.2 M NaOH are mixed. Then the P^{H} of the resulting solution is
a) 7 b) 1.097 c) 12.903 d) 11.903
- Among the following
1) On dilution, the P^{H} of an acid increases
2) A solution with $\text{P}^{\text{H}} = 5$ is 100 times more basic than a solution with $\text{P}^{\text{H}} = 3$
3) A solution with $\text{P}^{\text{H}} = 8$ is 1000 times more acidic than a solution with $\text{P}^{\text{H}} = 11$
4) The P^{H} of 10^{-9}M KOH is slightly greater than 7
a) All are correct b) 1, 4 are only correct c) 1,2,3 are only correct d) All are wrong
- What is the concentration of CN^- ions in a solution with 0.1 M HCl and 0.01 M HCN where K_a of HCN is 10^{-6} ?
a) 10^{-4} M b) 10^{-5} M c) 10^{-6} M d) 10^{-7} M
- pH of a solution of the mixture of 0.1N HCl and 0.1N CH_3COOH is ($K_a = 2 \times 10^{-5}$)
a) 1 b) 2 c) 1.7 d) 0.7
- Given $\text{HF} + \text{H}_2\text{O} \xrightleftharpoons{K_a} \text{H}_3\text{O}^+ + \text{F}^-$; $\text{F}^- + \text{H}_2\text{O} \xrightleftharpoons{K_b} \text{HF} + \text{OH}^-$. Which relation is correct :
a) $K_a \times K_b = K_w$ b) $K_b = 1/K_a$ c) $K_a = K_b = K_w$ d) $K_a / K_b = K_w$

More than One correct answer Type Questions

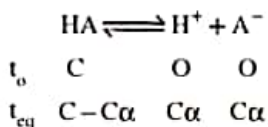
- pH of a weak acid in 0.1 M solution is 4.3. Which statements are correct ?
a) its 0.01M solution has $\text{pH} > 4.3$ b) its 0.01M solution has $\text{pH} < 4.3$
c) its 0.01M solution degree of dissociation is more than in 0.1 M solution
d) its 0.01M solution degree of dissociation is less than in 0.1 M solution

13. 100 ml of 0.1 M HCl and 100 ml of 0.1 M HOCN are mixed then ($K_a = 1.2 \times 10^{-6}$)
- OCN⁻ concentration in the solution is 1.2×10^{-6}
 - pH of the solution is 1.3
 - solution is a buffer
 - H⁺ in the solution is 10^{-6}

Linked Comprehension Type Questions

Passage :

Weak acids and bases are not completely ionised when dissolved in polar medium like water.



$$K_a = \frac{C\alpha^2}{1-\alpha} \approx C\alpha^2 ; \alpha = \sqrt{\frac{K_a}{C}} ; \therefore \frac{\alpha_1}{\alpha_2} = \sqrt{\frac{K_{a1}}{K_{a2}}} ; \frac{\alpha_1}{\alpha_2} = \sqrt{\frac{C_2}{C_1}}$$

(For two acids at same conc.) (for same acid at diff conc.)

14. α_1 and α_2 are in the ratio 1: 2 at same conc. $K_{a1} = 2 \times 10^{-4}$, what will be K_{a2} ?
- 8×10^{-4}
 - 2×10^{-4}
 - 4×10^{-4}
 - 1×10^{-4}
15. 0.01M CH₃COOH is 4.24% ionised. What will be the percentage ionisation of 0.1 M CH₃COOH.
- 1.33 %
 - 4.24 %
 - 5.24 %
 - 0.33%
16. Relative strength of two weak monoprotic acids may be given as
- $\frac{[\text{H}^+]_1}{[\text{H}^+]_2}$
 - $\frac{\alpha_1}{\alpha_2}$
 - $\sqrt{\frac{K_{a1}}{K_{a2}}}$
 - All of the above

Matrix Matching Type Questions

- | | |
|---|---|
| 17. Column-I (In H ₃ PO ₄ solution) | Column-II (C = [H ₃ PO ₄]) |
| A) [H ⁺] | P) $\sqrt{K_1 \cdot C}$ |
| B) [H ₂ PO ₄ ⁻] | Q) K ₂ |
| C) [HPO ₄ ²⁻] | R) $\sqrt{K_1 K_2}$ |
| D) [PO ₄ ³⁻] | S) $\frac{K_2 K_3}{\sqrt{K_1 \cdot C}}$ |

Integer type Questions

18. Calculate the change in pH of water when 0.01 mole of NaOH are added in 10 litre water.
19. Find the percentage degree of dissociation of 0.05M NH₃ at 25° C in a solution of pH = 11.
20. The degree of dissociation of 0.1M weak acid is 10⁻² and the degree of dissociation of the same acid in 0.025M concentration is 'x', then find 100x _____?

EXERCISE-III

(Buffer solutions, Buffer capacity, Salt hydrolysis, pH of amphoteric salts)

LEVEL-I (MAIN)

Straight Objective Type Questions

- Which of the following pair of solutions does not form a buffer solution ?
 1) NaH_2PO_4 and Na_2HPO_4 2) H_2CO_3 and NaHCO_3
 3) NH_4OH and NH_4Cl 4) KOH and K_2SO_4
- 1M NaCl and 1M HCl are present in an aqueous solution. The solution is
 1) Not a buffer solution with $\text{pH} < 7$ 2) Not a buffer solution with $\text{pH} > 7$
 3) A buffer solution with $\text{pH} < 7$ 4) A buffer solution with $\text{pH} = 7$
- A buffer solution contains a weak acid HA and A^- . When small quantity of NaOH is added, to keep pH as constant, which of the following reaction takes place?
 1) $\text{HA} \rightarrow \text{H}^+ + \text{A}^-$ 2) $\text{H}^+ + \text{A}^- \rightarrow \text{HA}$
 3) $\text{HA} + \text{OH}^- \rightarrow \text{H}_2\text{O} + \text{A}^-$ 4) $\text{A}^- + \text{H}_2\text{O} \rightarrow \text{HA} + \text{OH}^-$
- Which of the following is correct for buffer [Salt = S; acid = A]
 1) $\text{pK}_a = \text{pH} + \log \frac{[\text{S}]}{[\text{A}]}$ 2) $\text{pH} = \text{pK}_a + \log \frac{[\text{S}]}{[\text{A}]}$ 3) $[\text{H}^+] = 10^{-\text{pK}_a}$ 4) $\text{pH} = -\log \frac{[\text{S}]}{[\text{A}]}$
- On diluting a buffer solution, its pH :
 1) is increased
 2) is decreased
 3) remains constant
 4) will change which cannot be predicted unless dissociation of its weak acid (or base) is provided.
- For the buffer solution containing NH_4OH and NH_4Cl , pH of the buffer solution can be increased by
 1) Adding some more NH_4Cl 2) Adding some more NH_4OH
 3) Removing NH_4Cl 4) Both 2 and 3
- An acidic buffer contains equal concentrations of acid and salt. The dissociation constant of acid is 10^{-5} . The pH of the buffer solution is
 1) 5 2) 9 3) 4.49 4) 5.5
- K_a for HCN is 5×10^{-10} at 25°C . For maintaining a constant pH of 9, the volume of 5M KCN solution required to be added to 10ml of 2M HCN solution is ($\log 2 = 0.3$)
 1) 4ml 2) 8ml 3) 2ml 4) 10 ml
- pH of a mixture of HA and A^- buffer is 5. K_b of $\text{A}^- = 10^{-10}$. Hence $[\text{HA}]/[\text{A}^-]$ will be
 1) 1 2) 10 3) 0.1 4) 100
- A buffer solution contains 0.2mole of NH_4OH and 0.2 mole of NH_4Cl per litre. The pK_b of NH_4OH is 4.75. The pH of the buffer will be
 1) 4.75 2) 5.75 3) 9.25 4) 2.25

11. From the following table

Buffer solution	Vol. of 0.1M weak acid (in ml)	Vol. of 0.1M sodium salt of weak acid (in ml)
I	4.0	4.0
II	4.0	40.0
III	40.0	4.0
IV	0.1	10

Which of the two sets of buffer solutions have least pH ?

- 1) I & II 2) I & III 3) II & III 4) II & IV

12. Which buffer solution has maximum pH ?

- 1) mixture which is 0.1M in CH_3COOH and 0.1M is CH_3COONa [$\text{pK}_a(\text{CH}_3\text{COOH}) = 4.74$]
 2) mixture which is 0.2M CH_3COOH and 0.2M is CH_3COONa
 3) mixture which is 0.1M in NH_4Cl and 0.1M in NH_4OH [$\text{pK}_a(\text{NH}_4^+) = 9.26$]
 4) all the solutions have equal pH which is 4.74

13. Buffer capacity of acidic buffer solution is maximum when

- a) $\text{pH} = \text{pK}_a$ b) $[\text{salt}] = [\text{acid}]$ c) $\text{pK}_a = 7$ d) $[\text{H}^+] = \text{pK}_a$
 1) All are correct 2) b,c,d are correct
 3) a and b are correct 4) c and b are correct

14. When 0.02 mole of an alkali is added to an acid buffer, the pH changes from 4.75 to 4.8 the buffer capacity is

- 1) 2.5 2) 0.25 3) 0.4 4) 0.025

15. Aqueous solution of KCl is neutral because

- 1) K^+ undergoes hydrolysis 2) Cl^- undergoes hydrolysis
 3) Both K^+ and Cl^- undergo hydrolysis 4) No hydrolysis takes place

16. Assertion (A) : Aqueous solution of ZnSO_4 is neutral.

Reason (R) : Salt of strong acid and strong base does not undergo hydrolysis.

17. Assertion (A) : Aqueous solution of Na_2CO_3 shows $\text{pH} > 7$.

Reason (R) : Salt of strong base and weak acid undergoes anionic hydrolysis.

18. The no. of hydroxyl ions produced by one molecule of Na_2CO_3 on hydrolysis is

- 1) 4 2) 2 3) 3 4) 0

19. Nature of 0.1M solution of sodium bisulphate is

- 1) Acidic 2) Alkaline 3) Neutral 4) Amphoteric

20. In the titration of acetic acid versus sodium hydroxide, the pH of the solution at equivalence point (when temperature is 25°C) is :

- 1) about 5.5 2) about 6.5 3) about 7 4) about 8.5

21. The pH of an aqueous solution of NH_4CN (K_a of HCN is 9.2×10^{-10} & K_b of NH_4OH is 1.8×10^{-5})

- 1) > 7 2) < 7 3) 7 4) 14

22. The hydrolysis constant of CH_3COONa is given by

1) $K_h = \frac{K_w}{K_a}$ 2) $K_h = \frac{K_w}{K_b}$ 3) $K_h = \frac{K_w}{K_a \cdot K_b}$ 4) $K_h = K_a/K_b$

23. Hydrolysis constant of salt derived from strong acid and weak base is 2×10^{-5} . The dissociation constant of the weak base is

1) 5×10^{-8} 2) 5×10^{-9} 3) 5×10^{-10} 4) 2×10^{-19}

24. The hydrolysis constant of NaX (K_a of HX is 2×10^{-6}) is

1) 5×10^{-9} 2) 2×10^{-10} 3) 2×10^{-6} 4) 10^{-7}

25. The pH of 0.1M solution of the following compounds increases in the order

1) $\text{NaCl} < \text{NH}_4\text{Cl} < \text{NaCN} < \text{HCl}$ 2) $\text{HCl} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{NaCN}$
3) $\text{NaCN} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{HCl}$ 4) $\text{HCl} < \text{NaCl} < \text{NH}_4\text{Cl} < \text{NaCN}$

26. Assertion (A) : Aqueous solution of ammonium acetate is neutral.

Reason (R) : Dissociation constants of NH_4OH (K_b) and that of CH_3COOH (K_a) are nearly equal.

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- A buffer solution is prepared by mixing 50 mL of 0.2M of acetic acid with 'x' mL of 0.2M of NaOH solution. If pH of the resulting buffer solution is 4.7 then value of 'x' is ($K_a = 2 \times 10^{-5}$)
a) 28mL b) 22mL c) 24mL d) 25mL
- In the titration of a solution of weak acid, HX, with NaOH, the pH is 5.8 after 10mL of NaOH solution has been added and 6.40 after 20.0mL of the NaOH has been added. What is the ionisation constant of the acid HX ?
a) 7.94×10^{-7} b) 7.94×10^{-6} c) 7.94×10^{-8} d) 7.94×10^{-9}
- Which of the following is an acidic salt that can form an alkaline solution ?
a) NaHSO_4 b) NaHCO_3 c) Na_2HPO_3 d) NaCN
- If 50ml of 0.2M KOH is added to 40ml of 0.5M HCOOH . the pH of the resulting solution is ($K_a = 1.8 \times 10^{-4}$, $\log 18 = 1.26$)
a) 3.74 b) 5.64 c) 7.57 d) 3.42
- 1M NH_4OH and 1M HCl are mixed to make total volume of 300mL. If pH of the mixture is 9.26 and $\text{p}K_a(\text{NH}_4^+) = 9.26$ then volume ratio of NH_4OH and HCl will be :
a) 1:1 b) 1:2 c) 2:1 d) 3:1
- $\text{H}_2\text{O} + \text{H}_3\text{PO}_3 \rightleftharpoons \text{H}_3\text{O}^+ + \text{H}_2\text{PO}_3^-$, $\text{p}K_1 = x$
 $\text{H}_2\text{O} + \text{H}_2\text{PO}_3^- \rightleftharpoons \text{H}_3\text{O}^+ + \text{HPO}_3^{2-}$, $\text{p}K_2 = y$
Hence, pH of 0.01 M NaH_2PO_3 is :
a) $x + y$ b) $(x + y)/2$ c) $(x + y)^{1/2}$ d) $(x + y)^{1/3}$
- If the buffer capacity of a buffer solution is x, the volume of 1M NaOH added to 100mL of this solution to change the pH by 1 is
a) 0.1x mL b) 10x mL c) 100x mL d) x mL

More than One correct answer type questions

8. The incorrect statement(s) is/are
 a) CH_3COOH is a weak acid
 b) NH_4Cl gives an alkaline solution in water
 c) CH_3COONa gives an acidic solution in water
 d) NH_4OH is a strong base
9. Which of the following solution will have no effect on pH on dilution ?
 a) 0.3 M CH_3COOK
 b) 0.4 M $\text{CH}_3\text{COONH}_4$
 c) 0.2 M NH_4OH + 0.2 M NH_4Cl
 d) 0.3 H_2CO_3 + 0.3 M NaHCO_3
10. Which of the following mixtures can act as buffer?
 a) H_2CO_3 + KOH (2.5 : 1 molar ratio)
 b) H_2CO_3 + KOH (2 : 3 molar ratio)
 c) NH_4OH + HNO_3 (6 : 4 molar ratio)
 d) NH_4OH + HNO_3 (3 : 5 molar ratio)
11. The acid ionization hydrolysis constant of Zn^{2+} is 1.0×10^{-9} which of the following statement are incorrect?
 a) The $[\text{H}_3\text{O}^+]$ ion concentration in 0.001 M ZnCl_2 solution is 1.0×10^{-4}
 b) The pH of 0.001 M ZnCl_2 solution is 6
 c) The basic dissociation constant of $\text{Zn}(\text{OH})^+$ is 1.0×10^{-5}
 d) The basic dissociation constant of $\text{Zn}(\text{OH})^+$ is 1.0×10^5
12. Acetic acid and aq. NH_3 are weak monobasic acid and weak monoacidic base respectively and K_a of acetic acid is equal to K_b of aq. NH_3 . Which of the following statements are correct?
 a) All proportions of the above mixing would result in a neutral solution having pH = 7 at 25°C
 b) If aq. NH_3 is exactly half neutralized by HCl , then pOH of resulting solution is equal to $\text{p}K_b$
 c) If acetic acid is exactly half neutralized by NaOH , then pH of resulting solution is equal to $\text{p}K_a$
 d) If acetic acid is exactly neutralized by aq. NH_3 then pH of resulting solution is equal $1/2 \text{ p}K_w$

Linked Comprehension type questions**Passage-I :**

Higher the amount of acid or base used to produce a definite change of pH in a buffer solution, higher will be its buffer capacity. Buffer capacity of solution is maximum under the following conditions:

$[\text{Salt}] = [\text{Acid}]$ (in acid buffer) ; $[\text{Salt}] = [\text{Base}]$ (in base buffer)

pH of a buffer solution lies in the range given below: $\text{pH} = \text{p}K_a + 1$

In other words, any buffer solution can be used as buffer up to two pH units only, depending upon the value of $\text{p}K_a$ or $\text{p}K_b$. A buffer is said to be efficient when $\text{pH} = \text{p}K_a$ or $\text{pOH} = \text{p}K_b$.

13. Which among the following solution will be the most efficient buffer?
 a) 0.1M CH_3COONa + 0.01M CH_3COOH
 b) 0.1M NH_4Cl + 0.1M NH_4OH
 c) 0.0001M HCOOH + 0.002M HCOONa
 d) All of the above
14. The buffer capacity is equal to :
 a) $\frac{n}{\Delta\text{pH}}$
 b) $\frac{\Delta\text{pH}}{n}$
 c) $\text{p}K_a = \pm 1$
 d) none of these

15. Buffer capacity is maximum when:

- one mole of NH_4Cl is added to one mole of NH_4OH
- one mole of NH_4OH is added to one mole of HCl
- one mole of NH_4Cl is added to one mole of NaOH
- one mole of NaCl is added to one mole of NaOH

Passage-II :

A solution containing a weak acid and its conjugate acts as an acidic buffer. In a buffer the dissociation of the weak acid is suppressed by its conjugate base. (K_1, K_2 and K_3 of H_3PO_4 are 10^{-4} , 10^{-8} , 10^{-13} respectively)

16. What is the pH of a solution obtained by mixing 100ml of $0.1\text{M H}_3\text{PO}_4$ and 150ml of 0.1M NaOH .

- 8
- 9
- 7
- 6

17. Which of the following volume of 0.1 M NaOH added to 100 ml of $0.1\text{M H}_3\text{PO}_4$ does not form a buffer

- 50ml
- 150 ml
- 200 ml
- 250 ml

Matrix Matching Type Questions

18. **Column-I (mixed in equal volumes)**

Column-II (Nature of solution)

A) $0.2\text{ M HCN} + 0.1\text{ M KOH}$

P) Strong base

B) $0.2\text{ M HCl} + 0.1\text{ M KCN}$

Q) basic buffer

C) $0.2\text{ M NH}_3 + 0.1\text{ M HCl}$

R) acidic buffer

D) $0.2\text{ M KOH} + 0.1\text{ M NH}_4\text{Cl}$

S) Strong acid

19. Match the effect of addition of 0.1 M NaOH to 0.1 M , $50\text{ mL H}_3\text{PO}_4$ K_{a1} , K_{a2} , K_{a3} are the I, II, III ionization constant of H_3PO_4 respectively.

Column-I (Vol. of NaOH added)

Column-II (pH of resultant solution)

A) 150 mL of NaOH

P) $\text{pH} = \text{p}K_{a2}$

B) 75 mL of NaOH

Q) $\text{pH} = \text{p}K_{a1}$

C) 25 mL of NaOH

R) $\text{pH} = 7 + \left[\frac{\text{p}K_{a3} + \log c}{2} \right]$

D) 100 mL of NaOH

S) $\text{pH} = \frac{\text{p}K_{a2} + \text{p}K_{a1}}{2}$

Integer Type Questions

20. A certain buffer solution contains equal concentration of X^- and HX . K_b for X^- is 10^{-10} . Find the pH of buffer.

21. 0.15 mole of pyridinium chloride has been added into 500 cm^3 of 0.2 M pyridine solution. Calculate pH of the resulting solution, assuming no change in volume. (K_b for pyridine = $1.5 \times 10^{-9}\text{M}$)

22. Calculated the pH of solution prepared by mixing 40 ml of 0.1M acetic acid with 40 ml of 0.1M sodium acetate. Given $K_a = 10^{-5}$.

23. If the equilibrium constant for the reaction of weak acid HA with strong base is 10^9 , then calculate the pH of 0.1 M NaA .

OBJECTIVE CHEMISTRY IB**CHEMICAL EQUILIBRIUM**

24. In a titration, to 50 mL of 0.2 M of acetic acid ($K_a = 1.8 \times 10^{-5}$), 0.2 M of 'V' mL of NaOH is added to get a resultant solution of pH = 4.74. If $V = x^2$, what is x ?
25. 20 ml of 0.1 M NH_3 solution is titrated with 0.025M HCl solution. What is the pH of the reaction mixture at equivalence point at 25°C ? (K_b of NH_3 is 2×10^{-6}).
26. The dissociation constant of a substituted benzoic acid at 25°C is 1.0×10^{-4} . Find the pH of a 0.01M solution of its sodium salt.
27. If P^{K_a} of acetic acid and P^{K_b} of ammonium hydroxide are 4.76 each. Find the pH of ammonium acetate.

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL-I

- 1) 4 2) 2 3) 4 4) 4 5) 3 6) 4 7) 2 8) 2
 9) 4 10) 2 11) 4 12) 3 13) 3 14) 3 15) 4 16) 1
 17) 1

LEVEL-II

- 1) b 2) b 3) c 4) d 5) c 6) d 7) d 8) b
 9) b 10) d 11) d 12) c 13) bc 14) bc 15) bc 16) bcd
 17) acd 18) ab 19) abcd 20) A-T; B-P; C-QRS; D-R
 21) A-PQRS; B-Q; C-PQRS; D-Q 22) 5

EXERCISE-II

LEVEL-I

- 1) 4 2) 4 3) 4 4) 3 5) 2 6) 3 7) 3 8) 2
 9) 3 10) 4 11) 3 12) 3 13) 4 14) 4 15) 4 16) 3
 17) 1 18) 4 19) 4 20) 4 21) 4 22) 4 23) 3 24) 3
 25) 4 26) 1 27) 2

LEVEL-II

- 1) c 2) a 3) a 4) d 5) b 6) a 7) a 8) a
 9) d 10) a 11) a 12) ac 13) ab 14) a 15) a 16) d
 17) A-P; B-P; C-Q; D-S 18) 4 19) 2 20) 2

EXERCISE-III

LEVEL-I

- 1) 4 2) 1 3) 3 4) 2 5) 3 6) 4 7) 1 8) 3
 9) 3 10) 3 11) 2 12) 3 13) 3 14) 3 15) 4 16) 4
 17) 1 18) 2 19) 1 20) 4 21) 1 22) 1 23) 3 24) 1
 25) 2 26) 1

LEVEL-II

- 1) d 2) a 3) b 4) a 5) c 6) b 7) c 8) bcd
 9) bcd 10) abc 11) ad 12) bcd 13) b 14) a 15) a 16) a
 17) c 18) A-R; B-S; C-Q; D-P 19) A-R; B-P; C-Q; D-S 20) 4
 21) 5 22) 5 23) 9 24) 5 25) 5 26) 8 27) 7